

Exponential Distribution - $X \sim \text{Exponential}(\lambda)$

Used to model between the occurrence of events in an interval of time. i.e. Queuing Scenarios. Given parameter $\lambda > 0$,

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{if } x < 0. \end{cases}$$

$$F(x) = P(X \leq x) = 1 - e^{-\lambda x}, x \geq 0.$$

Note that Exponential distribution is **memoryless**.

$$P\{X \geq x + s \mid X \geq s\} = P\{X \geq x\}.$$

Expectation and Variance of $X \sim \text{Exponential}(\lambda)$

$$\mathbb{E}(X) = \frac{1}{\lambda} \quad \text{Var}(X) = \frac{1}{\lambda^2}.$$

Gamma Distribution- $X \sim \Gamma(\alpha, \lambda)$

The gamma density function depends on two parameters, $\alpha > 0$ and $\lambda > 0$ and is defined as

$$g(t) = \begin{cases} \frac{\lambda^\alpha}{\Gamma(\alpha)} t^{\alpha-1} e^{-\lambda t} & \text{if } t \geq 0 \\ 0 & \text{if } t < 0. \end{cases}$$

where

$$\Gamma(x) = \int_0^\infty u^{x-1} e^{-u} du, \quad x > 0$$

and $\Gamma(1) = 1$, $\Gamma(n) = (n-1)\Gamma(n-1)$. $\alpha = 1 \implies X \sim \text{Exponential}(\lambda)$.

Special cases: If n is a positive integer, $\Gamma(n) = n!$. $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$.

Expectation and Variance of $X \sim \Gamma(\alpha, \lambda)$

Sum of α independent exponential r.v.

$$\mathbb{E}(X) = \frac{\alpha}{\lambda} \quad \text{Var}(X) = \frac{\alpha}{\lambda^2}.$$

Suppose X is a random variable with density function $f_X(x)$ and Y is a random variable that can be expressed in terms of X , say, $Y = g(X)$ for some function g .

How to determine density function of Y (or $f_Y(y)$)?

- 1 Find the CDF of Y (or $F_Y(y)$), i.e., $P\{Y \leq y\}$.
(Express in terms of probability of event about X)
In other words, obtain $P(X \leq h(y))$.
- 2 Then, differentiate the CDF with respect to y .

Note: Do look out for the range of values of $y \in Y$ that satisfies the range of values of $x \in X$.

Proposition for strictly monotonic functions $g(X)$

Let X be a continuous random variable with density $f_X(x)$ and let $Y = g(X)$ where g is a **differentiable, strictly monotonic function** on some interval I , where $f_X(x) = 0$ if $x \notin I$. Then, Y has the density function

$$f_Y(y) = \begin{cases} f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right| & \text{if } y = g(x) \text{ for some } x \in I \\ 0 & \text{otherwise.} \end{cases}$$

where $g^{-1}(y)$ is the inverse function of g : $g^{-1}(y) = x$ if $y = g(x)$.

Example (Reference: Tutorial 5, Question 7)

Find the density of $Y = cX$, where $c > 0$ and $X \sim \text{Exponential}(\lambda)$.

Solution

$$\begin{aligned}F_Y(y) &= P(Y \leq y) = P(cX \leq y) \\&= P\left(X \leq \frac{y}{c}\right) \text{ as } c > 0 \\&= 1 - e^{-\left(\frac{y}{c}\right)\lambda} \text{ for } \frac{y}{c} \geq 0 \iff y \geq 0.\end{aligned}$$

Differentiating $P(Y \leq y)$ with respect to y ,

$$\begin{aligned}f_Y(y) &= \frac{d}{dy} [1 - e^{-\left(\frac{y}{c}\right)\lambda}] \\&= \frac{\lambda}{c} e^{-\left(\frac{\lambda}{c}\right)y} \text{ for } y \geq 0.\end{aligned}$$

Remark: You can also use the previous proposition, where $x = g^{-1}(y) = \frac{y}{c}$.